SCHEME OF VALUATION - QP code: 25901054
MG1SDCSDS102 - Mathematics for System & Software

Maximum Marks: 70 Marks **Time : 2 Hours**

PART A – Very Short Answers (Any 5 × 2 = 10 marks)

1. Check divisibility (2 marks)
   i) 3 ∤ 7 (7 not divisible by 3)
   ii) 3 | 12 (12 = 3×4)
   Award: 1 mark each
2. Relation between two sets (2 marks)
   $n(A \cup B) = n(A) + n(B) - n(A \cap B)$
   Example required.
   Award: 1 mark formula + 1 mark example
3. adj(A) (2 marks)
   Cofactor matrix is given → Take transpose.
   Award: 2 marks
4. Trace of matrix (2 marks)
   Definition: Sum of diagonal elements.
   Award: 1 mark definition + 1 mark example
5. Adjacent vertices (2 marks)
   Vertices with a connecting edge.
   Example must be shown.
   Award: 1+1
6. Graph definition (2 marks)
   G=(V,E). Example required.
   Award: 1+1
7. Boolean algebra operations (2 marks)
   AND, OR, NOT.
   Award: 2 marks
8. Distributive laws (2 marks)
   A(B+C)=AB+AC, A+BC=(A+B)(A+C).
   Award: 1 mark each law

PART B – Short Answers (Any 5 × 6 = 30 marks)

9. Verification of modular multiplication (6 marks)
   LHS: 13×7=91 → 91 mod 5 = 1.
   RHS: (13 mod 5)=3, (7 mod 5)=2 → 3×2=6 → 6 mod 5 = 1.
   Verified.
   Marks: Steps 4 + final 2
10. De Morgan's Laws proof (6 marks)
    State both laws and prove with membership table. $(A \cup B)^c = A^c \cap B^c$ and $(A \cap B)^c = A^c \cup B^c$
    Marks: Laws 2 + tables 4
11. Prove A adjA = |A| I (6 marks)
    Show multiplication gives determinant, |A|=-1 on diagonal.
    Marks: adjA 2 + multiplication 3 + conclusion 1
    ($\text{adj}A = \begin{bmatrix} 5 & -3 \\ -2 & 1 \end{bmatrix}$)
12. Idempotent matrix check (6 marks)
    Compute A^2 and compare with A. $A^2 \neq A$ → Not idempotent.
    Marks: computation 5 + conclusion 1
13. Handshaking + edges (6 marks)
    Sum of degrees = 2E.

10 vertices ×6 = 60 → E=30.
Marks: theorem 2 + working 3 + answer 1

14. Draw graphs C5, W4, K5, K1,5 (6 marks)
Each correct labelled diagram = 1½ marks.
4 diagrams × 1.5 = 6 marks

15. SOP expansion (6 marks)
F = xy + yz′ → Write minterms.
Marks: 3 + 3

16. F+G and FG (6 marks)
Truth table with 8 rows.
Column for F, G, F+G, FG.
Marks: table 5 + final 1

PART C – Essay Answers (Any 2 × 15 = 30 marks)

17. (i) Affine cipher (8 marks)
Use f(p)=2p+5 mod 26.
Convert text → apply function → convert back.
Marks: conversion 3 + encryption steps 3 + final text 2

(ii) Pseudorandom numbers (7 marks)
xn+1=(4xn+3) mod 4, start x0=3.
Generate 10 values.
Marks: 5 steps + 2 final sequence

18. Solve system by two methods (15 marks)
(i) Cramer (7 marks): Δ, Δx, Δy, compute x,y.
(ii) Inverse matrix (8 marks): adj(A), |A|, $A^{-1}B$.
Full stepwise marks

19. (a) Euler path/circuit (7 marks)
Definitions + criteria (even/odd degrees).
(b) Hamilton circuit in Kn (8 marks)
Proof Kn has Hamilton circuit for n≥3.
Marks based on clarity

20. Logic circuits (15 marks)
(i) AND gate explanation (3 marks)
(ii) Construct circuit for given output (6 marks)
(iii) Only OR + NOT for 3 variables (6 marks)